

## Circuit Analysis (Lab1)

### Experiment 1: Introduction to Lab Instruments and Components

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**Grade and Signature:** .....

|                              |                                                                                                                                 |                         |                             |                         |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------------------|-----------------------------|-------------------------|
| <b>CLO2</b>                  | Use with identification the circuit components, breadboards, multimeters, power supplies, signal generators, and oscilloscopes. |                         |                             |                         |
| <b>Psychomotor/Affective</b> | <b>Level1<br/>(1)</b>                                                                                                           | <b>Level2<br/>(2-3)</b> | <b>Level3<br/>(4-5)</b>     | <b>Level4<br/>(6-7)</b> |
|                              |                                                                                                                                 |                         |                             |                         |
| <b>Report Marks (3)</b>      |                                                                                                                                 |                         | <b>Total marks<br/>(10)</b> |                         |

**Objectives:** This experiment will provide exposure to the various test equipment to be used in subsequent experiments. A primary purpose of this lab course is for you to master the use of electronic test equipment.

**Equipment required:**

Power Supply, Breadboard, Oscilloscope, Signal Generator Connecting wires, Resistances, Multimeter.

**Procedure**

**BREADBOARD-BASICS:**

Breadboards (aka. Solderless board, Prototype board) are simply a set of pre-wired interconnected strips that are accessible through periodically spaced hole in the board. Looking at Figure 1. You can identify which holes form an interconnected strip by the black lines connecting them. By plugging the lead of a component into a hole you will be connected to all the other components in that strip without permanently connecting them. This allows you to build, alter, and test your prototype circuits quickly. There are two basic types of strips. The first type is called connection strip, they typically take up most of the board and are connected horizontally. Each hole can be uniquely identified using the labels “a-j” column labels and ‘1- 63’ row labels.

**NOTE:** ‘a-e’ connection strips are not connected to the ‘f-i’ connection strips. The second type is called bus strip. ALL the holes in a bus strip are connected vertically. Bus strips are typically labeled ‘A’ or ‘B’ and are marked by a red or blue line along their length.

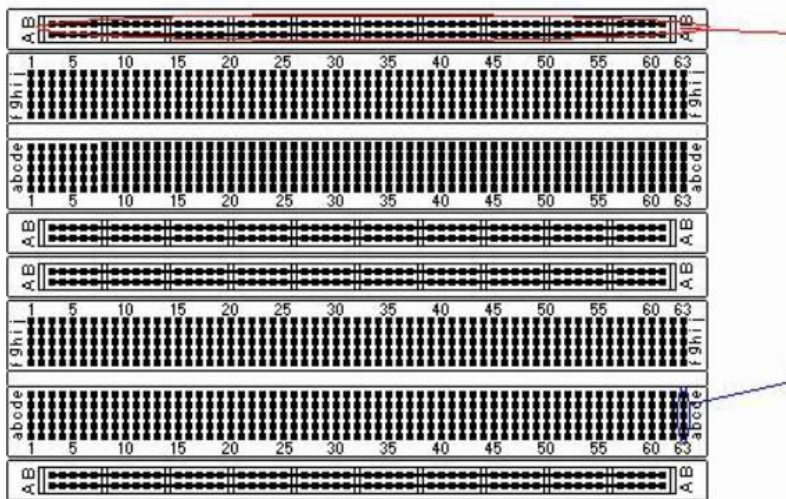


Figure 1 Breadboard

- This is an example of a Bus. A Bus is often used as way of delivering a common voltage to the entire system. In this case, breadboard. You may use a bus to make a common ground for the circuit you will build in this class.
- This is common strip

## Recognition of Basic Components

Here we are going to get acquainted to some of the basic components used in circuits such as the resistors. We will learn how to read the values using the codes as well as how to measure them using the ohmmeter.

### Resistors

You will be given some resistors, use the instruments to find their values and fill out the Table 1-1.

Table 1 Comparison between nominal and measured values.

|            | Measured   |
|------------|------------|
| Resistor 1 | 328.4 Ohms |
| Resistor 2 | 1.02k Ohms |
| Resistor 3 | 9.91k Ohm  |

## Recognition of Basic Equipment

Before proceeding with the lab, please familiarize yourself with setting up the bench equipment. Refer to Section 3 for details.

## OSCILLOSCOPE

Figure shows the digital storage oscilloscope.

### Features

#### Intensity Control

Rotate clockwise to increase the display intensity; counterclockwise to decrease. You can vary the intensity control to bring out signal detail, much like an analog oscilloscope. Digital channel waveform intensity is not adjustable.

#### Auto-scale Key

When you press the Auto scale key the oscilloscope will quickly determine which channels have activity, and it will turn these channels on and scale them to display the input signals.



Figure 2 Oscilloscope

### *Vertical Position Control*

Use this knob to change the channel's vertical position on the display. There is one Vertical Position control for each channel.

### *Channel On/Off Key*

Use this key to switch the channel on or off, or to access the channel's menu in the soft-keys. There is one Channel On/Off key for each channel.

### *Vertical Sensitivity*

Use this knob to change the vertical sensitivity (gain) of the channel.

### *Auto-Probe Interface*

When you connect a probe to the oscilloscope, the Auto Probe Interface attempts to determine the type of probe and set its parameters in the Probe menu accordingly.

### *Horizontal Delay Control*

When the oscilloscope is running, this control lets you set the acquisition window relative to the trigger point. When the oscilloscope is stopped, you can turn this knob to pan through the data horizontally. This lets you see the captured waveform before the trigger (turn the knob clockwise) or after the trigger (turn the knob counterclockwise).

### *Horizontal Sweep Speed Control*

Turn this knob to adjust the sweep speed. This will change the time per horizontal division on the display. When adjusted after the waveform has been acquired and the oscilloscope is stopped, this has the effect of stretching out or squeezing the waveform horizontally.

### *Measure Keys*

Press the Cursors key to switch on cursors that you can use for making measurements. Press the Quick Measure key to access a set of predefined measurements

### *Measurements using Oscilloscope*

The oscilloscope is primarily a voltage-measuring device. Once you have measured the voltage, other quantities are just a calculation away. For example, Ohm's law states that voltage between two points in a circuit equals the current times the resistance. From any two of these quantities, you can calculate the third. Another handy formula is the power law: the power of a DC signal equals the voltage times the current. Calculations are more complicated for AC signals, but the point here is that measuring the voltage is the first step towards calculating other quantities.

Ohm's Law:

$$\text{Voltage} = \text{Current} \times \text{Resistance}$$

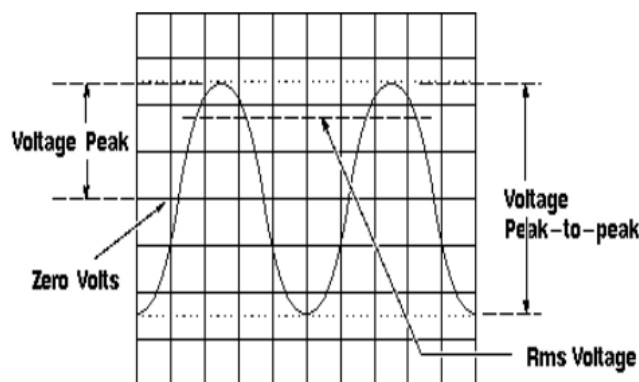
$$\text{Current} = \frac{\text{Voltage}}{\text{Resistance}}$$

$$\text{Resistance} = \frac{\text{Voltage}}{\text{Current}}$$

Power Law:

$$\text{Power} = \text{Voltage} \times \text{Current}$$

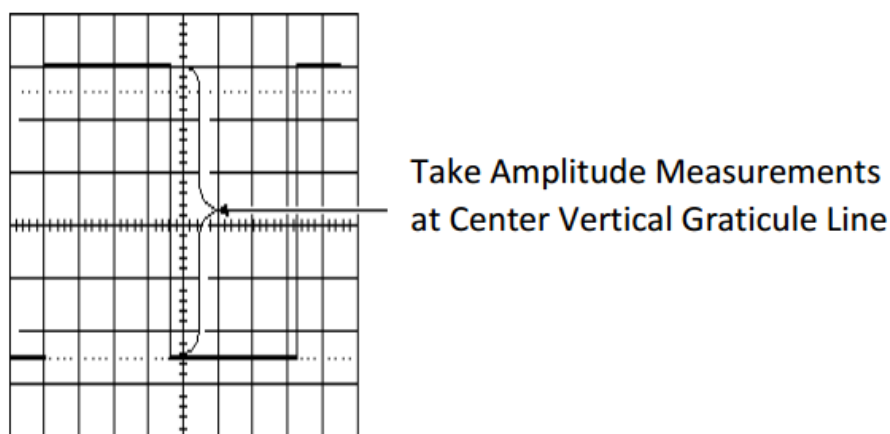
Following figure shows the voltage of one peak -  $V[p]$  - and the peak-to-peak voltage -  $V[p-p]$  -, which is usually twice  $V[p]$ . Use the RMS (root-mean-square) voltage -  $V[RMS]$  - to calculate the power of an AC signal.



**Figure 3 Voltage Peak and Peak-to-peak Voltage**

### Steps to Measure Voltage

1. Each probe has an alligator clip for connecting the scope to ground. Connect one of these clips to your circuit's ground. It's not necessary to connect the clip on the other probe.
2. Connect a probe to a point in your circuit that has a voltage you want to measure.
3. If there's another voltage you'd like to measure, connect the other probe to this second point.
4. Each square on the screen's grid is one centimeter. Above, you set the VOLT/DIV knobs to 1V. This means that each centimeter represents 1 volt, and each smaller division represents 0.2 volts.
5. If the voltage you're measuring is greater than 8 volts, the trace will go off the screen. If this happens, set the VOLT/DIV knob to a larger voltage.
6. An alternative to having the zero points for both traces on the bottom line of the grid, is to have one zero point on the center grid line, and the other on the bottom line. If the voltages you're working with go negative as well as positive, you'll probably want to have one zero point on the line 2 centimeters down from the top and the other on the line 2 centimeters up from the bottom.



**Figure 4 Measure Voltage on the Center Vertical Graticule Line**

### Frequency Measurement

Frequency can be measured on an oscilloscope by investigating the frequency spectrum of a signal on the screen and making a small calculation. Frequency is defined as the several times a cycle of an observed wave takes up in a second. The maximum frequency of a scope can measure may vary but it always in the 100's of MHz range. To check the performance of response of signals in a circuit, scope measures the rise and fall time of the wave.

### Method to Measure Frequency

- Increase the vertical sensitivity to get the clear picture of the wave on the screen without chopping any of its amplitude off.
- Now adjust the sweep rate in such a way that screen displays a more than one but less than two complete cycles of the wave.
- Now count the horizontal number of divisions of one complete cycle on the graticule from start to end.
- Now take horizontal sweep rate and multiply it with the number of units that you counted for a cycle. It will give you the period of the wave. The period is the number of seconds each repeating waveform takes. With the help of period, you can simply calculate the frequency in cycles per second (Hertz).

### Method to Measure Current

Attach a probe with the resistor to an electrical circuit. Make sure that resistor's power rating should be equal or greater than the power output of the system.

Now take the value of resistance and plug into Ohm's Law to calculate the current.

According to Ohm's Law,

$$\text{Current} = \frac{\text{voltage}}{\text{resistance}}$$

### Signal Generator

The signal generator is used to generate signals for your circuits. You will need to know how to set the signal generator to get sine, square, triangle or ramp signals. In addition, you will have to set up the frequency, the amplitude, offset voltage and the duty cycle. The default settings for this instrument are a sinewave of 1 kHz, with an amplitude of 100 mV and a DC offset of 0.0 V. The signal generator is very easy to use since each function has a specific button. If you want to select a waveform, just look for the button with the desired waveform such as a sine wave, a square wave, triangle wave, or ramp wave. Then, just press its button. All that you have to do now is set the parameters for the waveform. To set the frequency, amplitude, offset or the duty cycle you need to do the following:



**Figure 5 Signal Generator**

1. Press the appropriate gray buttons beneath the display screen (Freq/Period, Ampl/Hi Level, Offset/Lo Level, or Duty Cycle).
2. You may enter the value one of two ways.
  - a. Turn the knob and the highlighted digit will change. You may select a different digit by using the < or the > buttons.
  - b. You can also key in the digit by using number buttons.
3. Press "Output" button on the bottom right of the front panel (right next to Sync cable) and make sure the light is "on".

## Post Lab Questions

### 1. Define Oscilloscope and write its applications.

An oscilloscope is an electronic test instrument that graphically displays varying electrical voltages as a two-dimensional plot of one or more signals as a function of time. Applications include troubleshooting electronic circuits, measuring signal frequency and amplitude analyzing signal noise and observing wave shapes in fields like telecommunications, medical equipment testing, and engineering design.

### 2. What is Peak to Peak Voltages?

Peak to peak voltage is the difference between the maximum positive amplitude (peak) and the maximum negative amplitude (trough) of a waveform over one full cycle. It represents the absolute total vertical height of the wave.

### 3. What is RMS value?

Root Mean Square is the effective value of a varying voltage or current. It is defined as the equivalent direct current (DC) value that would produce the exact same amount of heat or power dissipation in a resistor as the alternating (AC) waveform.

### 4. Write the RMS values of Sine, Square and Triangular wave?

For a standard Sine wave the RMS value is found by dividing the peak voltage ( $V_m$ ) by the square root of 2, which is approximately 0.707 times the peak value. In the case of a Square wave the voltage remains at its peak level for the entire duration of the cycle meaning its RMS value is simply equal to the peak voltage ( $V_m$ ). For a Triangular wave, the voltage increases and decreases linearly, resulting in an RMS value of the peak voltage divided by the square root of 3, or roughly 0.577 times the peak value.

### 5. Explain DC offset?

DC offset is the mean amplitude displacement of a waveform from zero. It occurs when an alternating current (AC) signal is superimposed on a direct current (DC) baseline causing the entire wave to shift vertically up or down the Y-axis rather than being perfectly centered at zero volts.